Music-Driven Dance Generation for Simulated Humanoid Robots via Motion Retargeting

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***Abstract*—We propose a system that generates physically plausible humanoid robot dance motions directly from music input. Our pipeline consists of three stages: (1) generating human dance motion sequences from music using the pre-trained diffusion model EDGE, (2) retargeting the generated SMPL-format motions to a simulated humanoid robot in MuJoCo through joint mapping and motion adaptation, and (3) executing and evaluating the retargeted motions under physics simulation. We compare two retargeting strategies—direct SMPL-shaped humanoid mapping and cross-embodiment retargeting to the Unitree G1 robot—and evaluate the system across multiple music genres using beat alignment score, physical stability rate, and joint limit violation metrics. To our knowledge, this work represents one of the first attempts to bridge *learning-based, diffusion-model* music-conditioned dance generation with physics-based humanoid robot simulation in an end-to-end pipeline.**

I. Motivation and Problem Statement

This is an **application-oriented research project** that bridges two active but largely disconnected research areas: music-conditioned dance generation and physics-based humanoid robot control.

Recent advances in generative models have enabled high-quality 3D human dance synthesis from music [10, 8, 2]. However, these methods produce kinematic motion sequences on the SMPL human body model [5] without considering physical feasibility. The generated motions may contain poses that violate joint limits, lose balance, or produce physically impossible accelerations—none of which matter for animation rendering, but all of which are critical for robotics applications.

On the other side, physics-based humanoid control has made remarkable progress through reinforcement learning [7], yet these systems typically track pre-recorded motion capture data rather than algorithmically generated content.

Our key question is: Can we close the loop from music input to physically plausible robot dance execution in simulation? This requires solving the additional challenges of cross-embodiment motion retargeting and physical stability under the constraints of a simulated robot body.

This problem is meaningful for several reasons: (1) it tests the physical plausibility of state-of-the-art dance generation models, (2) it explores cross-embodiment motion transfer between human body models and robots, and (3) it has potential applications in entertainment robotics and human-robot interaction.

II. Literature Review

Music-conditioned dance generation. Li et al. [4] introduced the AIST++ dataset and the FACT model as a baseline for music-to-dance generation. Siyao et al. [8] proposed Bailando, which uses a VQ-VAE to encode dance motions into discrete tokens and an Actor-Critic GPT for music-conditioned autoregressive generation. Tseng et al. [10] introduced EDGE, a diffusion-based model that generates editable dance sequences conditioned on Jukebox music features, achieving state-of-the-art quality and supporting in-painting for flexible editing. More recently, Li et al. [2] proposed FineDance with body-part-level generation, and Li et al. [3] introduced Lodge for long-sequence dance generation through a coarse-to-fine diffusion framework.

Physics-based humanoid control. Luo et al. [7] proposed Perpetual Humanoid Control (PHC), which trains a universal RL policy to imitate arbitrary motion capture sequences on an SMPL-shaped humanoid in MuJoCo [9], achieving high success rates on the AMASS dataset (with its improved PHC+ variant further closing the gap on challenging motions). Luo [6] released SMPLSim, a tool for generating MuJoCo-compatible SMPL humanoid models. For cross-embodiment transfer, Araújo et al. [1] proposed GMR for retargeting human motions to diverse humanoid robots (e.g., Unitree H1/G1) in real time.

Research gap. Existing dance generation works focus on kinematic quality without physical validation. Existing humanoid control works track motion capture data, not algorithmically generated dances. No prior work has connected music-conditioned dance generation models with physics-based humanoid simulation as an end-to-end pipeline.

III. Technical Plan

Our system consists of three main modules, as illustrated in Figure 1.

Stage 1: Music-conditioned dance generation. We use the pre-trained EDGE model [10] to generate dance motion sequences from music. EDGE employs a Transformer-based diffusion model conditioned on Jukebox music features (4800-dim per frame) and outputs SMPL motion representations ($\mathbf{m} \in \mathbb{R}^{K\times 151}$, comprising 24-joint 6D rotations, root translation, and foot-ground contact labels). Jukebox features are pre-computed on a cloud GPU due to memory requirements ($\geq$16GB VRAM). The diffusion model generates 5-second clips that are stitched for longer sequences via an overlapping

Music (.wav)
↓ *Jukebox feature extraction*
EDGE Diffusion Model
↓ *SMPL 24-joint rotations + root*
Motion Retargeting
(A) SMPLSim direct mapping | (B) GMR → Unitree G1
↓ *Robot joint angles*
MuJoCo Physics Simulation
↓ *PD / PHC tracking control*
Evaluation & Rendering

Fig. 1. System pipeline overview.

in-betweening scheme. If EDGE proves infeasible, Bailando [8] serves as a lightweight alternative using librosa audio features.

Stage 2: Motion retargeting. We investigate two retargeting strategies:

- **Strategy A (SMPL-to-SMPL):** Using SMPLSim [6] to generate an SMPL-shaped MuJoCo humanoid. Although the joint topology matches EDGE's output, retargeting still requires converting rotation representations (SMPL 6D/axis-angle to MuJoCo quaternion), aligning rest poses between the kinematic and physics models, and selecting appropriate body shape parameters that affect the humanoid's mass distribution and limb lengths.
- **Strategy B (SMPL-to-Robot):** Using GMR [1] to retarget motions to the Unitree G1 robot model (37 actuated DoFs, including simplified hand joints) from MuJoCo Menagerie. This requires solving the correspondence between SMPL's 24 joints and G1's different kinematic structure, handling joint limit differences, and adapting motion amplitude.

Stage 3: Physics simulation and control. Retargeted joint trajectories are tracked in MuJoCo [9] using a progressive control strategy. We begin with PD controllers for simplicity and interpretability, which are expected to work for slow and stable motions (e.g., gentle swaying, simple steps). For more dynamic dance sequences where PD control is likely insufficient due to balance and contact challenges, we escalate to the RL-based motion tracking policy from PHC [7], which has demonstrated robust tracking on diverse motion capture sequences. We monitor physical stability by tracking the center-of-mass height (fall detection), joint torques, and ground contact forces.

IV. Expected Outcomes

1) An end-to-end system: music input → simulated robot dance video.
2) Quantitative evaluation across multiple music genres (e.g., hip-hop, ballet, house) using: **Beat Alignment Score** [4] (measuring the alignment between motion kinematic beats and music beat positions via a Gaussian-kernel metric; we adopt EDGE's variant using acceleration peaks as motion beats), **Physical Stability Rate** (fraction of frames without falling), and **Joint Limit Violation Rate**.
3) Comparative analysis of two retargeting strategies (direct SMPL mapping vs. cross-embodiment to Unitree G1).
4) Analysis of which dance genres are more physically feasible for robots (e.g., gentle ballet vs. aggressive hip-hop).
5) Demo videos showcasing the full pipeline.

V. Potential Risks and Backup Plan

Risk	Prob.	Backup Plan
Jukebox extraction exceeds GPU memory	Med	Switch to Bailando (uses librosa, CPU-compatible)
Robot frequently falls in simulation	High	(1) Reduce motion amplitude via scaling factors; (2) restrict to slow-tempo genres (e.g., lyrical, waltz); (3) if still insufficient, fall back to kinematic playback for visualization
GMR retargeting to G1 produces poor results	Med	Use Strategy A only (SMPLSim direct mapping)
EDGE environment setup fails	Low	Use Bailando with simpler dependencies

VI. Timeline

Phase	Milestone
Phase 1	Environment setup (EDGE + MuJoCo + SMPLSim)
Phase 2	Dance generation with pre-trained EDGE
Phase 3	Motion retargeting (Strategy A) + sim playback
Phase 4	PD control + PHC RL tracking + stability experiments
Phase 5	Strategy B (G1 retargeting) + multi-genre eval
Phase 6	Quantitative evaluation + final report + demo

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